Course Syllabus

Nanoscale Systems in Chemical Engineering

Department of Chemical Engineering, University of Illinois - Chicago

Spring 2024, Monday/Wednesday from 10:30 AM - 11:45 AM, building EIB room 271

Zoom link:

https://uic.zoom.us/j/9335717023?pwd=VGxRbFZ3SlM0Ly83VmZ6YXV5TU9adz09

Required Textbooks and Materials

Required Texts:

"Introduction to Nanoscience", S.M. LINDSAY. ISBN-10: 0199544212

Suggested Course Materials

Suggested Readings/Texts :

1. Introduction to Nanomaterials and Devices by Omar manasreh.
2. Introduction to Nanoscience and Nanotechnology, Chris Binns
3. Nanomaterials and Nanochemistry**,** by Bréchignac, C., Houdy, P., Lahmani, M

Professor Contact Information

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Office Hours : By appointments

Other Information:

Course Pre-requisites:

GOALS AND OBJECTIVES or Course Description:

Nanoscale science and nanotechnology have many potential applications and an enormous economic impact. Nanotechnology is related to the ability to manipulate and engineer physical, chemical, biological, and optical properties that emerge naturally at the nanoscale. The objective of this course is to introduce basic concepts and modern methods in major developments of nanotechnology and nanomaterials. Introduces tools and principles relevant at the nanoscale materials. Discusses current and future nanotechnology applications in engineering, materials, physics, chemistry, electronics, and energy.

Student Learning Outcomes:

1. Explain the history of nanotechnology.
2. Understand the nanoscale paradigm in terms of properties at the nanoscale dimension.
3. Demonstrate a working knowledge of nanoscience/nanotechnology principles and applications
4. Apply key concepts in materials science, chemistry, physics, biology, and engineering to the field of nanotechnology.
5. You will be expected to demonstrate what you have learned by problem solving in homework/exams and by giving a presentation to the class.
6. The student will be able to gain the ability to oral and written communicate effectively.

Lectures Notes:

The lectures will be used to introduce key concepts as well as new materials. PPT and chalk/blackboard will be both used. PPT slides will be posted to Blackboard after the class. Students are encouraged to take their own notes.

Course Schedule or Course Contents

I. Introduction

1. History of Nanotechnology
2. What is Nanoscience?
3. Size Matters
4. Nanometers, Micrometers, Millimeters–Visualizing a Nanometer

II. Basic/Fundamentals

1. **Application of Quantum Mechanics to Nanomaterial Structures**

2. **Statistical mechanics and chemical kinetics**

III. Tools

Microscopy and manipulation tools (Student Present Projects).

1. The scanning tunneling microscope
2. The atomic force microscope
3. Electron microscopy
4. Nano-measurement techniques based on fluorescence
5. Nanotechnology for Chemical Engineers:
6. Chemical Vapor Deposition (CVD) Techniques in Nanotechnology (Final presentaion)
7. Chemical Vapor Synthesis (CVS) of Nanostructures

IV. Applications

Electrons in nanostructures

1. The vast variation in the electronic properties of materials
2. Electrons in nanostructures and quantum effects
3. Drude model
4. Fermi gas Fermi liquids and the free electron model
5. Electrons in crystalline solids: Bloch's theorem
6. *Electron in* 3D materials.
7. *Electron in 2D materials*
8. *Electron in 1D materials (nanowires or Quantum Well)*
9. *Electron in 0D materials.* Quantum Dots Systems
10. Transport regimes in nanostructures: the Landauer resistance.
11. Single-Electron Transistors: The Coulomb blockade

Molecular electronics

1. Why molecular electronics?
2. Lewis structures as a simple guide to chemical bonding
3. The variational approach to calculating molecular orbitals
4. The hydrogen molecular ion revisited
5. Hybridization of atomic orbital
6. Electron transfer between molecules—the Marcus theory
7. Charge transport in weakly interacting molecular solids—hopping conductance
8. Single molecule electronics
9. Measurement and fabrication
10. Control of Electron Transport Through a Single Molecule
11. Density-Functional Theory for the Calculation of Molecular Electronics Properties

Magnetism in Nanostructures/Nanomaterials.

1. The Importance of Magnetism
2. Origin of magnetism (macro- and microscopic view)
3. Types of Magnetism (diamagnetic, paramagnetic, feri-, Anti-, fero-, magnetic)
4. Quantum spin models
5. Magnetism of Clusters (Ferromagnetic-antiferromagnetic particles)
6. Ferromagnetic particles becoming single domain
7. Superparamagnetism in small ferromagnetic particles
8. Nanodisks
9. Nanorings
10. Giant Magnetoresistance effect in hybrids (layered structures)
11. Applications of magnetic nanoparticles

OPTICAL PROPERTIES of Nanomaterials

Optical Properties: The interaction of electromagnetic radiation with matter

Size effect on optical properties

Spectroscopic Techniques for Studying Optical Properties of Nanomaterials

Optical Properties of Metal Nanomaterials: Plasmons, **Surface plasmons.** Plasmons in Nanostructures.

Optical properties of semiconductors : Semiconductor nanoparticles & films

Photoluminescence : Bulk, quantum wells, quantum wires and quantum dots

Fluorescence from core-shell quantum dot

Optical Properties of Composite Nanostructures

Metamaterials (with negative refractive index)

Applications of Optical Properties of Nanomaterials

Guest lectures:

1. X-ray technique to study nanomaterials: Tao Li (Northern Illinois University/ANL)
2. Mechanical properties of nanomaterials: Pallab Barain (Argonne National Laboratory)
3. Understanding the Energy Storage Principles of Nanomaterials in Lithium-Ion Battery: Zonghai Chen (Argonne National Laboratory)
4. Topics? Yi Cui (Stanford University)/ Larry Curtiss (ANL)

Grading Policy

Quizzes + Homework 40 %

Exam 1: 20 %

Presentation: 20%: The final project aims for students to study and present relevant topics currently being researched in modern nanotechnology applications.

Exam 2: 20 %